Learning and Evaluating Judges of Formal Specification Coverage

Anonymous submission

Abstract

An artificial intelligence (AI) judge can rank formalizations like human raters yet assign systematically different ratings or miss disagreement about individual artifacts. Correlation alone does not determine whether a judge predicts the human rating distribution for the same artifact. We study judges of formal specification coverage: how fully candidate theorem statements express the properties required by a reference. Each judge predicts six rating probabilities from Python source, a candidate Lean artifact, and a reference. We specify matched correlation-training and proper-loss-training conditions, selected by a common validation ranked probability score. Historical evidence covers 75 artifacts from 25 tasks, with 345 recorded ratings across five raters, including 195 from the three-expert subset. We separate that previously exposed evidence from an ongoing human-rating collection and a prospective study of task, solver and rater sample sizes. The resulting study tests judge–human agreement with explicit data provenance, distribution-sensitive metrics and limits on transfer; fresh comparative results remain pending.

1 Introduction

A formalization can prove every theorem it states and still omit a required property of the source program. Assessing theorem coverage therefore requires comparing the candidate statements with the intended obligations, not merely checking proof completion. We call this property *formal specification coverage* (FSC). The judge is the object of this paper: given a source program, candidate formalization and reference, how well does it predict human ratings? VeriBench provides the application and historical artifacts for this question (Miranda et al. 2026). Its broader code-verification benchmark is related work rather than this paper's contribution.

Large language model (LLM) judges offer a practical way to compare outputs with instructions and references. Zheng et al. (2023) study agreement with human preferences and document position, verbosity and self-enhancement biases. Such agreement depends on the task, rating procedure and judge configuration. In our historical FSC data, strong rank correlation coexists with a positive mean rating bias (Section 6). A ranking statistic alone cannot identify the human rating distribution for an individual artifact. This observation motivates a richer target; it does not establish that correlation training performs worse than loss training.

We model human ratings as a conditional distribution over the six categories zero through five. The judge receives the original Python source, candidate Lean artifact and reference shown to the raters. It predicts a probability for each rating rather than only a single scalar rating. Keeping individual ratings preserves information about disagreement that a consensus label would discard. Proper scoring rules provide losses whose expected optimum is the target distribution (Gneiting and Raftery 2007). Finite data, restricted judge classes and optimization can still prevent a fitted judge from attaining that optimum.

We specify a comparison between training with Pearson correlation and training with a proper probability loss. Both conditions use the same judge class, training data, computational budget and validation criterion. Normalized ranked probability score (RPS) is the primary training loss for the proper-loss condition and the common validation criterion. Cross-entropy and Brier loss define additional, separately reported training conditions. The fixed judges share a held-out evaluation panel with RPS primary and cross-entropy and total variation distance secondary. Pearson correlation measures linear association of output ratings; optimizing it does not require a linear judge.

The study has three connected parts: historical judge evidence, new human-rating collection, and estimation of the required collection size. The historical pool contains 25 tasks with three candidate samples each, yielding 75 artifacts. Five rater columns contain 345 observed ratings out of 375 possible; the primary three-expert subset contains 195 out of 225. The three candidates are samples, not three solver configurations. The new collection prepares versioned candidate packets and records each person's rating under a common rubric. Its main-round judge comparison is pending, and the final larger-study allocation remains open. The historical results motivate these choices without supplying independent validation for the new judge.

The intended outcome is a judge whose predictive error and transfer limits can be measured against a specified human-rating target. Human agreement remains distinct from

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

semantic truth and from a checked implication proof. We make the following contributions:

1. We define an artifact-conditional ordinal prediction task and a matched comparison of correlation training with proper-loss training.
2. We consolidate historical judge evidence with its rating counts, input provenance, uncertainty and prior-exposure limitations.
3. We specify how the ongoing collection and sample-size study will support a prospective comparison without treating planned ratings as completed evidence.

2 Target and Measurement Conditions

For task t and solver configuration a, let

$$z_{ta} = (x_t, \widehat{y}_{ta}, y_t^*), \qquad R_{tah} \in \{0, 1, 2, 3, 4, 5\}. \tag{1}$$

Here x_t is the original Python source and task requirements, $\widehat{y}_{ta}$ is the candidate Lean artifact, y_t^* is the reference, and h identifies a human rater. The normalized scalar rating is $R_{tah}/5$. The solver configuration includes the model, harness, tools, prompt, effort and generation budget. If a configuration produces multiple candidates for a task, an additional generation index g gives R_{tagh}. The historical three-candidate axis indexes generation samples, not three distinct solver configurations. The main study displays the complete candidate and reference artifacts to raters and the judge, including any proof bodies or `sorry` placeholders. Both receive identical input contents; the rendered view is recorded, and proof completion is a separate variable rather than the coverage target.

2.1 What coverage means

A formal specification is a theorem statement considered without its proof. Coverage concerns the required statements and the domains and assumptions under which they hold. A candidate may cover a reference statement with a stronger statement or with several statements together. Counting theorem names or measuring textual overlap cannot establish this implication. Conversely, narrowing a precondition can remove required cases even when a theorem's conclusion is unchanged. Proof completion is recorded separately from statement coverage.

The source program constrains the intended behavior but does not automatically resolve a conflict between implementation behavior, task requirements, and the reference. The protocol records such conflicts for independent inspection rather than silently choosing one input as infallible. Conditioning on a reference makes the rating task more concrete; it can also make both humans and a judge prefer the reference's representation. Reference sensitivity is therefore a measurement condition to test, not an assumed benefit of adding the reference.

2.2 The human distribution

Let $\mathcal{H}$ denote a specified population of qualified Lean and formal-methods experts using the same rubric. The target is the probability mass function

$$p^*(r \mid z) = \Pr_{h \sim \mathcal{H}}\left(R_h(z) = r\right). \tag{2}$$

This definition requires a recruitment frame and a rating procedure, including allowed assistance, time guidance, and treatment of ambiguous inputs. The ongoing collection requires localized AI assistance, followed by the human rater's own comparison and final rating. Historical instructions prohibited AI assistance, but compliance cannot be inferred where no disclosure was collected. Assistance conditions are therefore part of the target distribution and must accompany comparisons across rating rounds. A convenience sample does not automatically represent all experts. The empirical distribution from a fixed recruited panel is a different, narrower target; both must be labeled when reported.

The judge predicts $q_\theta(r \mid z)$, a nonnegative vector of six probabilities summing to one. The initial interface elicits those six probabilities explicitly as structured output; it does not infer them from token likelihoods or the frequency of repeated scalar answers. Any subsequently fitted probability mapping is trained on training ratings and selected on validation ratings, with its family and constraints recorded as a separate candidate condition. Its expected normalized rating is $\mu_q(z) = \sum_{r=0}^{5}(r/5)q_\theta(r \mid z)$. Each individual human rating is retained, including disagreement. A histogram of stochastic scalar judge outputs does not by itself represent the human distribution: model-call instability and human disagreement have different sources. The proposed probability-vector interface is consequently a new judge condition, not a reinterpretation of an old scalar score.

3 A Distribution-Sensitive Objective

Proper scoring rules reward accurate probability forecasts in expectation (Gneiting and Raftery 2007). For the cumulative probabilities $Q_c = \sum_{r=0}^{c} q_r$, we use the loss

$$S(q, r) = \frac{1}{5}\sum_{c=0}^{4}\big(Q_c - \mathbf{1}\{r \le c\}\big)^2. \tag{3}$$

The factor $1/5$ places RPS on $[0, 1]$. The ordered thresholds retain the ordinal structure: probability placed many categories away from an observed rating incurs more cumulative error than probability placed in an adjacent category. Equal threshold weights are an explicit design choice, not a claim that every practical consequence of a rating is equally spaced.

Let $P_c = \Pr(R \le c \mid z)$. For each threshold, expansion of the squared error gives

$$\mathbb{E}[S(q, R) \mid z] = D(q, p^*) + U(p^*), \tag{4}$$

$$D(q, p^*) = \frac{1}{5}\sum_{c=0}^{4}(Q_c - P_c)^2, \tag{5}$$

$$U(p^*) = \frac{1}{5}\sum_{c=0}^{4} P_c(1 - P_c). \tag{6}$$

The first term measures conditional distribution error; the second is the loss remaining even for a perfect forecast of a disagreeing expert population. Because the cumulative probabilities determine all six category probabilities, the unrestricted expected-loss minimizer is uniquely $q = p^*$. This

fact justifies the population training target. It does not guarantee that a particular model class, prompt search, or finite sample will attain that target.

For equal category spacing on $[0, 1]$, the one-dimensional Wasserstein distance is

$$W_1(q, p^*) = \frac{1}{5} \sum_{c=0}^{4} |Q_c - P_c|. \tag{7}$$

Cauchy–Schwarz gives $W_1(q, p^*)^2 \leq D(q, p^*)$. Thus small excess expected RPS bounds an interpretable distribution distance. More generally, for a normalized L-Lipschitz utility function f, the expectation gap is at most LW_1. This connects a distributional claim to uses such as estimating the mean rating, without requiring a separate match of every moment.

3.1 Why correlation or a pooled divergence is insufficient

Consider human scores $(0, 0.25, 0.5)$ and judge scores $(0.5, 0.75, 1)$. Pearson and rank correlations are one, although every score differs by 0.5. Alternatively, permuting predictions among artifacts preserves the marginal score histogram while destroying itemwise agreement. A divergence between pooled histograms misses that failure. The required comparison is conditional on the same artifact, then averaged across artifacts.

Logarithmic loss is another proper objective and connects to Kullback–Leibler divergence, but it ignores category order and diverges when the judge assigns zero probability to an observed category. With few ratings per artifact, directly estimating an entire human probability vector and plugging it into a divergence is also noisy. RPS can instead score each individual rating without declaring a sparse empirical histogram to be the true human distribution. We use cross-entropy as a separate training condition and a shared secondary evaluation metric, with the probability-floor policy specified before evaluation.

3.2 Training objectives and artifact weighting

For n artifacts, with assigned rater set $\mathcal{H}_i$ of size m_i, define

$$\widehat{L}_S(\theta) = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{m_i} \sum_{h \in \mathcal{H}_i} S(q_\theta(z_i), R_{ih}). \tag{8}$$

Artifacts receiving additional ratings retain the same total weight as other artifacts. Let $\bar{r}_i$ be the mean normalized human rating for artifact i. The correlation-training condition maximizes Pearson correlation between $\mu_q(z_i)$ and $\bar{r}_i$ across training artifacts. The proper-loss-training condition minimizes Equation 8 on the same training artifacts, with RPS primary. The judge class, inputs, data access and search budget are matched between conditions. Pearson correlation constrains the relationship between output ratings; the mapping from source artifacts to predictions may be nonlinear. Constant predictions or targets make correlation undefined and require a prespecified failure policy.

Additional proper-loss conditions train with cross-entropy or six-category Brier loss and remain separately identified. Cross-entropy scores an individual rating as $-\log q_r$ using natural logarithms. The six-category Brier loss is $\sum_{k=0}^{5}(q_k - \mathbf{1}\{r = k\})^2$, with range $[0, 2]$. It differs from the binary Brier loss used for a separate prover-outcome target. No population optimality result proves that one training procedure will perform better with finite data and a restricted judge class.

Every condition selects hyperparameters by the same validation RPS. Changing the validation criterion from correlation to proper loss would answer a different question from changing the training objective. All prompt changes, fitted mappings and attempted candidates are recorded because repeated selection can overfit validation data (Cawley and Talbot 2010). Final test ratings do not enter training or selection.

3.3 Shared evaluation metrics

RPS is the primary held-out metric for every training condition. Cross-entropy and total variation distance (TVD) are shared secondary metrics. For an artifact's empirical human distribution $\widehat{p}_i$, define

$$\widehat{\mathrm{TVD}} = \frac{1}{n} \sum_{i} \frac{1}{2} \sum_{r=0}^{5} |q_\theta(r \mid z_i) - \widehat{p}_i(r)|. \tag{9}$$

This TVD lies in $[0, 1]$ and weights artifacts equally. It is a finite-rating distribution diagnostic, not a proper individual-rating loss or total-variation mutual information. The empirical human distribution remains noisy when few people rate an artifact. Expected-rating correlations, absolute error, signed bias and human–human agreement supplement the common loss panel.

3.4 A worked distinction between uncertainty sources

Suppose, illustratively, that half of the qualified raters assign category zero to one artifact and half assign category five. The correct human forecast places probability $1/2$ on each endpoint, so every cumulative probability P_c equals $1/2$ for $c = 0, \ldots, 4$. Its expected RPS is 0.25, entirely due to human disagreement. A judge that always assigns all probability to category zero has expected RPS 0.50: its additional distribution error is 0.25. Both forecasts can be perfectly reproducible across repeated judge calls. The deterministic endpoint forecast is therefore stable but worse at predicting the human distribution.

These numbers are an exact illustrative calculation, not observed study results. They show why a low model-repeat standard deviation cannot replace the proper-loss comparison and why a positive raw RPS cannot by itself imply a defective judge. The same distinction applies when interpreting the dense subset: additional raters estimate the human cumulative probabilities more precisely, while additional judge calls estimate the variability of the probability forecast. The two replication budgets answer different questions and must be reported separately.

4 Human Data and Prospective Study Design

The study separates an existing rating pool, an ongoing collection of new ratings, and planning for a larger comparison.

Each record has its own candidate provenance, rater population, assistance conditions and exposure history. Reanalyzing or exporting an existing rating does not create a new observation.

4.1 Historical rating pool

The audited historical snapshot has 25 tasks, three candidate samples per task and five rater columns. This gives 375 possible ratings across 75 artifacts, of which 345 are observed. The designated three-expert subset contains 195 of 225 possible ratings: 45 artifacts have three expert ratings and 30 have two. Missing ratings are absent observations, not zeros. Section 6 reports the associated judge studies and their prior-exposure limitations. Older exports used different prompt passes and filtering rules; their counts are not silently substituted for this pinned snapshot.

4.2 Ongoing packet preparation and collection

The preserved main packet contains 25 tasks and five model configurations using the Terminus 2 harness, giving 125 candidate artifacts. Its collection target is two independent qualified ratings per artifact, or 250 main ratings; author ratings are additional. The checked collection record dated 11-11-2026 contains no completed main ratings. Warm-up ratings assess readiness for that task and do not count as main ratings.

A versioned replacement packet was built around Claude Code, Codex and Pi, with fewer Terminus 2 artifacts retained for comparison. Its plan called for 85 artifacts on 25 tasks: 25 each from native Codex, native Claude Code and Pi, and ten from Terminus 2. Only 54 planned artifacts were returned: 25 from Codex, 19 from Claude Code and ten from Terminus 2. The other 31, all 25 from Pi and six from Claude Code, are recorded as missing outputs; a missing output is not a human rating of zero. The replacement packet therefore contains 54 artifacts, and its target is two independent ratings per artifact, or 108 ratings; author ratings are additional. This is a collection workload, not collected data or a statistical sufficiency claim. No main rating of the replacement packet had been completed at the 11-11-2026 check, and most rater assignments remain pending. Because missing outputs differ by harness, the returned artifacts are not a random sample of each configuration's attempts. The ten-artifact Terminus 2 subset is exploratory; historical and new generation conditions differ, so it does not isolate a causal harness effect. The preserved packet's 125 artifacts and 250-rating target do not establish the replacement's size or completion. A harness name alone does not identify a solver configuration: exact models, tools, effort, prompts, budgets and source hashes must accompany each candidate. Existing outputs qualify for reuse only after their task inputs and generation conditions are checked against the intended comparison.

Raters compare the source, candidate and reference using the same coverage rubric. The current form records the final rating, AI-assistance and human-rating attestation, review time, confidence, rationale and problem notes. Localized AI assistance is required prospectively; the human makes the final comparison and rating. Independent submissions are preserved rather than replaced with a consensus answer. The inherited collection is reserved for evaluation in the new training-objective comparison, with prior exposure retained in its record.

4.3 Estimating tasks, configurations and raters

Let T count task identities, A solver configurations and H distinct raters. The observed number of ratings is $\sum_i m_i$, not TAH unless every rater rates every artifact. Generation repeats require their own index. Size planning compares variance-based precision calculations, learning-theory bounds and fitted learning curves under explicit assumptions. None of the recovered conversation estimates establishes a sufficient final allocation. Task-family dependence, reused raters and fixed solver configurations prevent treating all ratings as independent observations.

Earlier allocations remain archived planning scenarios, not measured requirements or adopted sizes for the replacement packet. A final allocation must be tied to a declared estimand, assignment pattern and precision target.

4.4 Identity separation and execution decisions

The prospective comparison requires a new assignment that separates training, validation and test task families, solver identities and rater identities. The earlier design's shared training/validation raters do not implement that strict three-way separation. The historical data also reuse raters and include prior test exposure; relabeling their splits cannot remove it. The amendment must define eligible identities and the exact transfer claim before new training or test use. In a leave-one-configuration-out analysis, that configuration's artifacts, ratings and examples are excluded from both training and validation for its fit. Such an analysis tests transfer of a fitting procedure to a specified configuration, not performance on every future agent.

A separately eligible training pilot can measure review time, disagreement, missingness and variability of paired losses. The final sample size must follow a declared precision or useful-effect target and a feasible time and cost budget. Before test inspection, fix the candidate manifest, rating procedure, assignment, judge versions, repeat count, analysis and failure policy. Trace input bytes through human display, judge input, probability parsing and scoring using hashes. A known-rating reproduction check tests that pipeline separately from semantic performance. These unresolved decisions are part of the study plan; they are not completed data collection or empirical validation.

5 Precision, Selection, and Decision Rules

The three dimensions of the study address different uncertainties. More distinct tasks cover a broader set of inputs and increase the precision of task-averaged comparisons. More solver configurations probe different output distributions. More distinct raters probe variation between raters, whereas more ratings of one artifact sharpen information about that artifact's rating distribution. The product of these counts is not the number of independent observations.

5.1 A transparent planning calculation

As an illustration, suppose the human-specific contribution to normalized loss has standard deviation 0.10. For indepen-

dent, representative raters, the normal-approximation 95% half-width of that contribution would be

$$1.96\frac{0.10}{\sqrt{H}} = \begin{cases} 0.080, & H = 6, \\ 0.049, & H = 16. \end{cases} \tag{10}$$

This is a hypothetical component of uncertainty, not an observed total interval. If repeated ratings share a human-specific effect, thousands of ratings by the same few raters do not remove it. The calculation motivates recruiting more distinct raters, while the actual allocation must be checked against pilot variance and workload.

Increasing ratings per artifact from three to eight reduces an independent mean's standard-error factor by $\sqrt{3/8} \approx 0.61$. That is a 39% reduction under the stated independence and common-variance approximation, not a guaranteed 39% reduction in the final study interval. Balanced reuse of raters creates cross-artifact dependence, and an exact finite-panel analysis differs from independent population sampling. These conditions motivate considering a dense subset alongside broad task coverage, subject to the final allocation.

For correlations, Fisher's approximation for independent bivariate-normal pairs is

$$\mathrm{SE}(\operatorname{atanh}\widehat{\rho}) \approx 1/\sqrt{n-3}. \tag{11}$$

In the actual study, multiple outputs share a task and ratings share a rater; substituting the total number of rating entries for n in that formula is invalid. Nor can additional data be expected to raise the true correlation: it improves estimation of whatever association exists. We therefore size the study against paired proper-loss precision and expert variation, and report correlation intervals under a dependence-aware analysis.

5.2 Estimating distribution error

The raw RPS includes the disagreement term in Equation 6. For a fixed artifact and threshold, let $X_h = \mathbf{1}\{R_h \le c\}$ and let s_X^2 be the sample variance with denominator $m-1$. Under independent draws from the specified expert population,

$$\widehat{D}_c = (Q_c - \overline{X})^2 - \frac{s_X^2}{m} \tag{12}$$

is unbiased for $(Q_c - P_c)^2$. The subtraction removes the variance of the empirical human cumulative probability. We average across thresholds and then artifacts. Individual corrected estimates can be negative and are not clipped before aggregation. Their intervals, rather than the sign of an individual estimate, determine what distribution error remains compatible with the data.

If the estimand instead concerns a fixed panel of H raters sampled uniformly without replacement, the subtraction is $(1 - m/H)s_X^2/m$. That correction targets the finite-panel mean, not the wider population automatically. More complicated assignment and nonresponse require corresponding design-based or model-based adjustments. A dense subset could make this diagnostic more informative but cannot remove uncertainty about unrepresented experts.

5.3 Uncertainty and comparison

Crossed task and rater identities require inference that respects both sources of dependence. Methods for resampling crossed arrays motivate independent reweighting of identities under appropriate assumptions (Owen and Eckles 2012). The exact sparse-assignment procedure, coverage checks, and treatment of small rater counts must be specified using pilot data before the test is inspected. We report fixed-panel analyses separately from analyses extending to an expert population. Task resampling estimates variation over tasks like those in the benchmark; for the complete fixed benchmark it is not uncertainty caused by missing benchmark task identities. Solver configurations are treated as fixed in the primary analysis unless a different sampling population is explicitly justified.

Primary comparisons use paired, artifact-weighted RPS differences between a constant training-estimated distribution, a fixed rubric/reference-assisted judge, a correlation-trained judge, and an RPS-trained judge, all evaluated on the same test artifacts. We also include a non-LLM theorem-count baseline to test whether a semantic model improves on a simple structural cue. Let $v = \min(1, N_{\text{candidate}}/N_{\text{reference}})$ count theorem statements without inspecting their proofs. For $u = 5v$, place probability $1 - (u - \lfloor u \rfloor)$ at $\lfloor u \rfloor$ and the remaining mass at the next category, with all mass at five when $u = 5$. This defines an ordinal probability forecast for comparison; it does not assert that counts establish coverage. A reference with no countable statements is flagged as an invalid baseline input rather than silently assigned a score. The declaration-counting rule is fixed in the pilot. Each probabilistic condition must produce the same six-category output format. Converting a historical scalar judge into a probability forecast is a new baseline condition and must be documented. Matched artifacts and rating assignments reduce unnecessary comparison noise. Unequal model calls, tokens, and runtime must be reported rather than hidden as equivalent budgets.

The constant forecast tests whether input-dependent predictions add information beyond the training rating frequencies. The count baseline tests whether a cheap structural signal explains the apparent gain. The fixed-rubric condition tests the benefit of fitting, while the two trained conditions test the choice of training objective. The trained conditions use the same judge class, information access and computational budget, then select hyperparameters by the same validation RPS. Separate cross-entropy and Brier training conditions use that same selection and evaluation procedure. Otherwise, an observed difference could reflect search resources or a different model family rather than the objective.

The pilot must set a minimum meaningful improvement, a tolerable distribution error, repeat-instability and failure limits, and the corresponding interval rule. A claim of success on every included configuration requires inference that accounts for those simultaneous comparisons. These thresholds are currently unset; a statistically nonzero improvement alone is not the proposed definition of practical success. Later changes after test inspection require fresh evidence or an explicitly exploratory label.

Statistic	Estimate	95% interval
Spearman ρ	0.826	[0.316, 0.966]
Kendall τ_b	0.698	[0.216, 0.913]
Pearson r	0.858	[0.707, 0.941]
ICC(A,1)	0.727	[0.391, 0.890]

Table 1: Historical selected ensemble: 30 artifacts from ten test tasks. Intervals are percentile task bootstraps with 10,000 requested draws. ICC(A,1) is single-score absolute-agreement intraclass correlation. The split had prior exposure; these are not fresh-study results.

6 Historical Judge Evidence

The historical archive contains several distinct analyses that must not be pooled or described as one original result.

6.1 Original audit and score mapping

The original VeriBench report describes a raw Codex command-line rubric audit on 75 artifacts (Miranda et al. 2026). It reports Pearson correlation 0.6137, Spearman correlation 0.3755 and Kendall correlation 0.2996 against its historical pooled human target. Two-fold isotonic score mapping in that audit gives 0.2108, 0.0461 and 0.0416, respectively. The source provides no confidence intervals for these entries. That target pools seven rating files, including two earlier-prompt passes from one person, and differs from the later five-rater snapshot. This full-file rubric audit is separate from VeriBench's deployed coverage scorer; its agreement cannot be transferred to a different judge implementation.

An early analysis reports Pearson correlation 0.703349 on 72 scored artifacts from a 75-artifact, 25-task pool. It uses a GPT-5.3-Codex judge with an isotonic score mapping fitted on training folds, against the mean of one expert's two earlier-prompt passes. Its Spearman and Kendall values are 0.321143 and 0.265060. These are two-fold task-split validation results without a separate test counterpart; the aggregate receipt does not supply confidence intervals.

6.2 Selected scalar judge and agreement limits

A later selected ensemble, source experiment 100, uses GPT-6 Sol with high reasoning effort and Claude Opus 5.5 with maximum compute effort through the historical Vals model gateway. For each member it takes the median of three calls, then averages the two medians: six calls per completed scalar score. It uses human rating instructions, training-derived conventions, and training examples. The 25 tasks are split 10/5/10 into training, validation, and test, retaining each task's three candidates together. The human target is the mean of available ratings from a three-expert panel, divided by five. Of the 30 test artifacts, 18 have three expert ratings and 12 have two.

The headline correlations in Table 1 coexist with normalized mean absolute error 0.201111 and mean bias $+0.181111$, with judge mean 0.653333 and human mean 0.472222. Bias is the judge rating minus the expert mean, so a positive value indicates greater leniency. A separately labeled post hoc task bootstrap of the frozen historical data gives 95% intervals [0.130, 0.273] for absolute error and [0.099, 0.262] for bias. It uses 10,000 draws and reproduces the original point estimates; it is new analysis of old data, not new human evidence. No new significance test is asserted for these quantities.

Excluding two test tasks whose references were placeholders when rated leaves 24 artifacts from eight tasks. On that ordinary-reference subset, Spearman correlation is 0.652 with interval $[-0.166, 0.916]$, and Pearson correlation is 0.835 with interval [0.454, 0.958]. The mean bias rises to $+0.205556$, from judge mean 0.566667 and human mean 0.361111. The subset correlation intervals use 1,998 nondegenerate draws out of 2,000 requested task bootstraps. Reference quality therefore changes both the interpretation and the uncertainty of the headline association.

The reported mean repeat standard deviation, 0.020488, also needs its original unit. It is computed from three per-pass averages, each combining one call from each model. It is not the standard deviation of repeated executions of the complete six-call median-based ensemble. The archive supplies no interval for that repeat statistic.

For context, the archived leave-one-expert-out comparison reports Spearman correlation 0.764 over all 75 artifacts and 0.747 over the 45 validation-plus-test artifacts of experiment 100. These compare one expert with the other available experts and have no uncertainty intervals in the imported aggregate. Their task sets and targets differ from the 30-artifact test comparison in Table 1; they cannot establish that the selected judge exceeds human agreement on a matched test set.

Several limitations prevent clean transfer claims. Candidate-generation records trace 74 of 75 files to three single-call, zero-shot samples per task from Claude 3.7 Sonnet at temperature 0.7, with one source receipt missing. These are repeated samples of one older model, not three different coding agents. The source receipt discloses prior exposure to the test ratings in earlier experiments and an initial descriptive audit; the exact total number of exposed judge versions and resulting selection inflation are not established by the inspected records. Nine references changed statement content between rating and earlier judging, although experiment 100 restored rating-era inputs. The reconstruction remains ambiguous for five tasks whose references changed during the rating window; six test artifacts from two tasks carry that ambiguity flag. The source appendix also reports three test items whose rating sheet linked a different candidate artifact. Excluding either group raises test agreement in the source analysis, but does not repair the original evidence or justify replacing its full denominator. Neither adding raters to those files nor narrowing their intervals would erase these provenance limits.

Finally, a separate five-fold automated recipe, source experiment 104, reports tie-preserving Spearman correlation 0.603 [0.287, 0.836] over all 75 artifacts and 25 tasks, with Kendall 0.498 and Pearson 0.621. Its original floating-point calculation gave Spearman 0.601533; the published addendum preserves mathematical ties before ranking. This evaluates fold-specific recipe fits, not the frozen selected ensemble. The archive retains both versions and their labels. These

original analyses concern scalar LLM judges; they do not evaluate the proposed native probability interface or validate a prover judge.

6.3 Retrospective comparison of training objectives

A subsequent matched comparison fits mixtures of five cached judge configurations on the same historical ratings. Three scalar repeats per configuration and artifact are converted to floored six-category histograms; no new semantic-model calls are made for this comparison. Both training objectives use the same mixture class and select shrinkage by validation RPS. Its original task split yields 30 training, 15 validation and 30 test artifacts, with 78, 39 and 78 observed expert ratings.

On the 30 test artifacts from ten tasks and nine task families, the normalized RPS difference, RPS-trained minus Pearson-trained, is $+0.005241$. The 95% family-bootstrap interval is $[-0.068074, +0.014864]$, using 1,998 valid refit/reselection draws out of 2,000 requested. Lower RPS is better; the point estimate favors Pearson, but the interval does not resolve an advantage or establish equivalence. Validation selected the RPS-trained mixture, and this selection was not changed after test inspection. No significance test is asserted.

The bootstrap refits and reselects mixtures but conditions on the cached model calls, prompts and rater identities. It retains historical task-family overlap and prior exposure rather than repairing them. Additional cached-metric, cross-validation and expanded-judge-pool analyses reuse this human evidence; a separate synthetic study contains no human observations. They motivate the prospective comparison but do not constitute independent replications or fresh-rating validation.

7 Semantic Controls and Reporting

Agreement with experts using a faulty reference can reproduce the same error rather than measure coverage correctly. We therefore accompany the random audit with controls whose semantic relationship is checked independently wherever possible. These controls do not receive the random audit's sampling interpretation, and their counts and selection process are reported separately.

Removing a necessary obligation should reduce coverage, provided no remaining statement already implies it. Narrowing a precondition should reduce coverage when it excludes inputs required by the task. Equivalent formulations should receive similar ratings, while adding irrelevant or vacuous statements should not increase meaningful coverage. Changing proof bodies without changing statements should preserve a statement-coverage rating. An incomplete or false reference should trigger discrepancy reporting where the evidence supports it. The exact tolerances and control counts are pilot decisions, not assumed measured properties of the judge.

For equivalence controls, compilation of two programs is insufficient. The relevant implications must preserve domains, assumptions, and obligation weights. For negative controls, a counterexample should establish that the removed obligation is not already entailed. AI can assist in constructing these certificates, but checker acceptance and the assumptions of the certificate must be recorded. Unresolved cases remain unresolved rather than being converted into negative examples by a timeout.

The reference-anchoring study is a separate controlled comparison within this research program. It varies reference representation and proof visibility while holding the intended obligations and candidate artifact fixed. Its labels, estimand, and pilot counts must not be substituted for those of the main conditional-rating study. Likewise, the human-rating study does not change an existing benchmark's prover-based scoring implementation.

7.1 What the final report must contain

The primary table will report RPS, paired baseline differences and confidence intervals on the frozen test set and for each included solver configuration. Cross-entropy and per-artifact TVD are reported on those same artifacts as secondary metrics. Any excluded-solver fits have their own transfer table, denominators and training-data eligibility record. The final assignment manifest determines these counts; earlier budget scenarios do not fix them. Distribution-error diagnostics on the dense subset, means, signed bias, absolute error, Pearson correlation, Spearman correlation, and Kendall correlation supplement these results. Human–human agreement is computed on matched artifacts, and the prediction mean is not confused with the full probability vector.

A separate repeatability panel reports variation across repeated probability vectors and expected scores, the number of full judge executions, and the aggregation policy. A stable but wrong judge can pass a repeatability check, while a correct prediction of human disagreement can intentionally be broad. The paper must keep these statements separate. Missing and failed calls appear with their original denominator; because RPS is bounded, worst- and best-case completion bounds can show whether missingness could reverse a conclusion.

Public evidence includes the protocol, configurations, distributable input manifests, analysis code and authorized results. The collection preserves authorized named ratings in its versioned repository record, with stable identifiers for matching responses and preventing duplicate counts. Private contact information, consent correspondence and sealed test data remain restricted. The anonymous submission reports aggregate evidence without participant identities or identifying repository links. Raw ratings retain the original values, missingness and assistance disclosures; a normalized analysis export is a separate derivative. The historical import preserves immutable aggregate receipts and hashes, with post hoc calculations labeled separately.

8 Related Work

VeriBench and formal specification coverage. VeriBench evaluates end-to-end coding and formal-verification agents, including whether their formal statements express required properties (Miranda et al. 2026).

Its tasks, reference artifacts and earlier human-rating study provide this paper's application context. The broader solver benchmark, proof-success comparisons and operational coverage scorer remain separate contributions. This paper extracts the human-rating judge question and preserves each historical scorer's input conditions, target and analysis. Predicting expert coverage ratings is also different from forecasting acceptance by a bounded proof-search procedure: a search failure does not establish non-entailment.

Language-model judges. Human-agreement studies evaluate language-model judges for conversational preferences and report biases associated with position, verbosity and self-preference (Zheng et al. 2023). Our task instead concerns formal statements, assumptions and reference coverage. Agreement statistics remain useful, but an ordinal probability forecast additionally makes disagreement and distributional error explicit. We compare training objectives with a common validation criterion rather than assume that any objective is empirically superior.

Scoring, selection and dependent observations. Proper scoring rules motivate probability forecasts whose expected score is optimized by the target distribution (Gneiting and Raftery 2007). Repeated model selection can bias subsequent performance assessment (Cawley and Talbot 2010), motivating separate validation and sealed test use. Crossed-array resampling addresses dependence induced by shared identities (Owen and Eckles 2012); its suitability still depends on the actual assignment and target population. These results guide the protocol but do not supply a universal number of tasks, solvers or raters.

9 Discussion and Current Status

The study asks which training objective yields the best practical judge of human ratings under matched resources. Proper scoring rules identify a distributional population target, but finite data, model restrictions and optimization can alter the empirical comparison. The historical mixture analysis leaves the RPS-versus-Pearson advantage unresolved. The selected scalar judge also shows why strong rank association can coexist with systematic leniency. Neither result establishes a winner for a future native-probability judge.

The three workstreams have different roles: historical ratings support retrospective analyses, the ongoing packet prepares new human evidence, and size planning determines what additional evidence is feasible. The replacement packet contains the 54 artifacts returned from 85 planned and targets 108 independent ratings; those ratings and the prospective split amendment remain unfinished. Exact judge pins, model-repeat counts, practical success thresholds and the sparse-assignment uncertainty procedure must be fixed before test inspection. Warm-up responses are collection progress, not completed main-round evaluation.

A successful study would support a bounded claim about predicting ratings under specified inputs, assistance conditions and rater recruitment. It would not equate agreement with semantic truth, establish every artifact's correctness, or guarantee transfer to unrepresented tasks, solvers or people. Independent semantic controls and checked proof evidence address complementary questions. The paper therefore contributes a defined comparison and traceable historical evidence while reserving fresh empirical conclusions for the new collection.

References

Cawley, G. C.; and Talbot, N. L. C. 2010. On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation. *Journal of Machine Learning Research*, 11: 2079–2107.

Gneiting, T.; and Raftery, A. E. 2007. Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association*, 102(477): 359–378.

Miranda, B.; Daruru, S.; Hersch, E. S.; et al. 2026. VeriBench: End-to-End Formal Verification Benchmark for AI Coding Agents in Lean 4. Preprint; historical judge analysis in Appendix G.

Owen, A. B.; and Eckles, D. 2012. Bootstrapping Data Arrays of Arbitrary Order. *The Annals of Applied Statistics*, 6(3): 895–927.

Zheng, L.; Chiang, W.-L.; Sheng, Y.; Zhuang, S.; Wu, Z.; Zhuang, Y.; Lin, Z.; Li, Z.; Li, D.; Xing, E. P.; Zhang, H.; Gonzalez, J. E.; and Stoica, I. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *Advances in Neural Information Processing Systems*, 36.